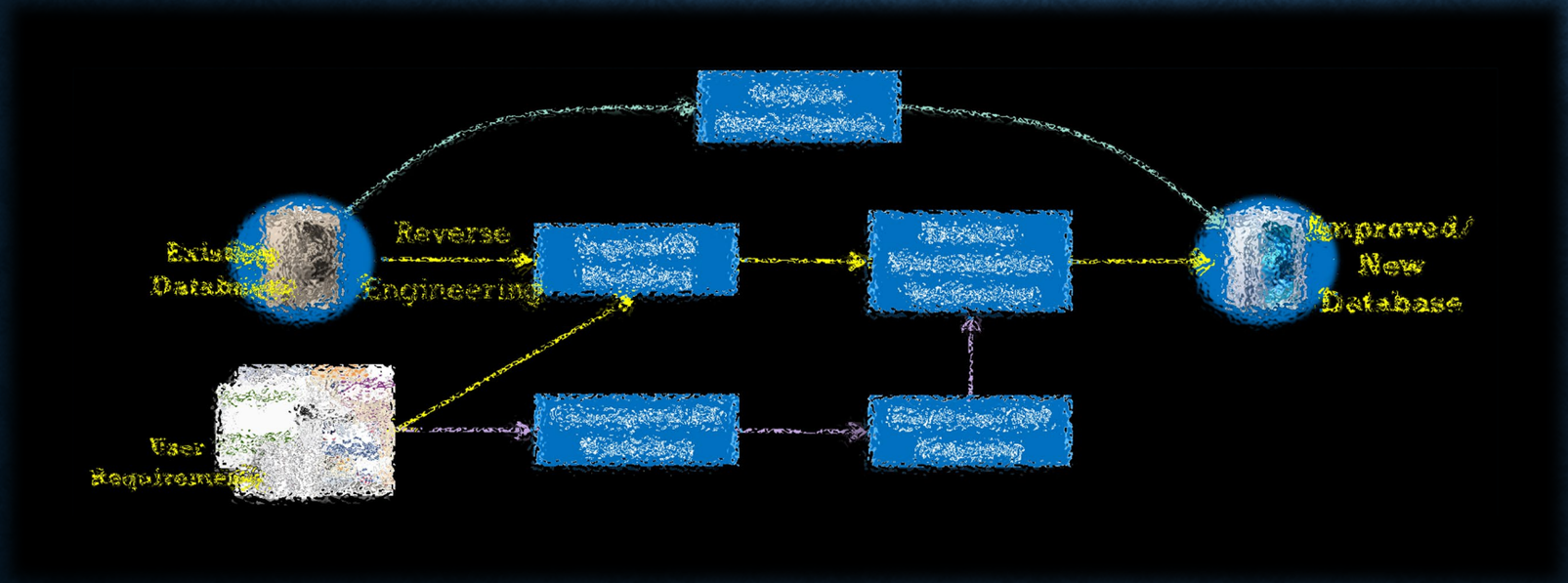
Relational Database & SQL

Relational DB Design Approach



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Computer Information Systems, Georgia State University

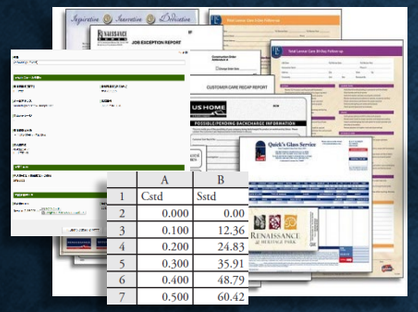
Relation Normalization Approach



- Assumes the current database sufficiently supports business needs.
- Focuses on structural improvements to
  - Resolve issues with data redundancy and inconsistency (update anomalies)
  - Modify schemas as necessary to adapt to evolving **user requirements**

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Reverse engineering approach



Updated
User Requirements

ER Model Modification

Relation Normalization Verification

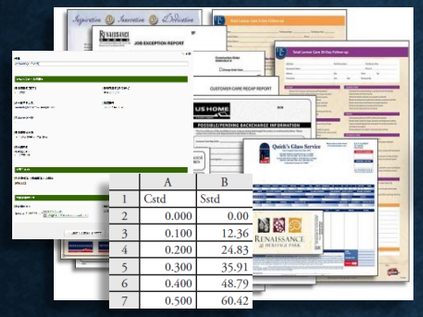
Logical ER Modeling

Improved Database

Current Database

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A new database design approach



User Requirements

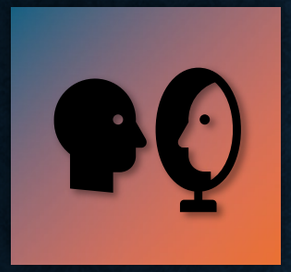
Conceptual ER Modelling

Relational DM Mapping

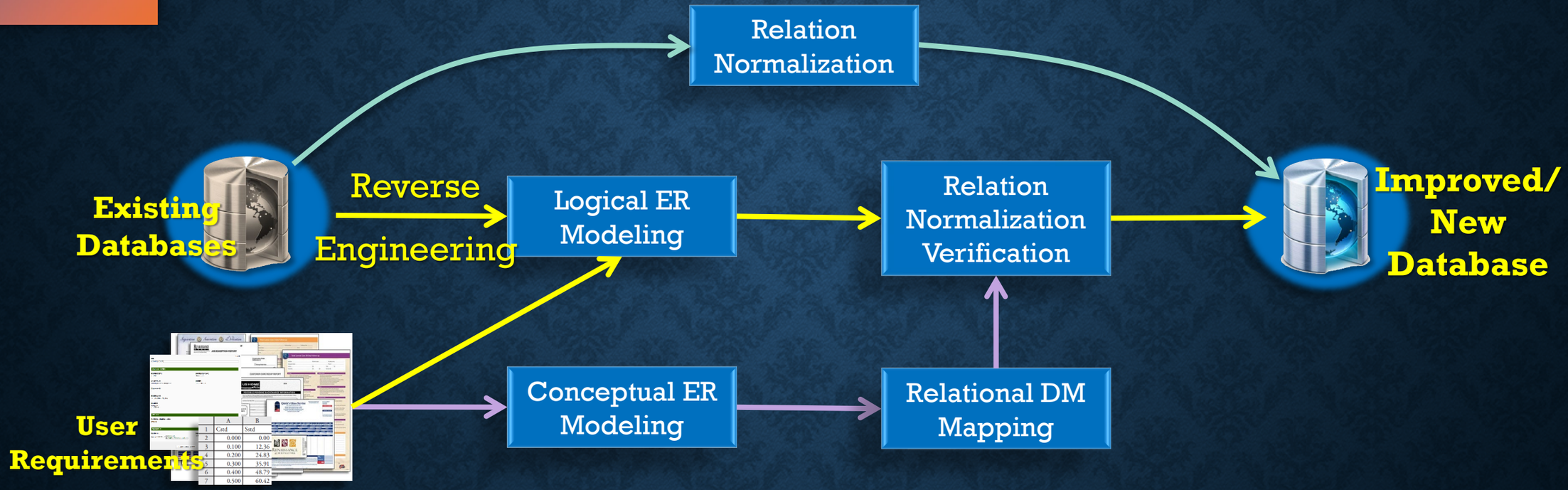
Relation Normalization Verification

DB New Database

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TAKEAWAYS…



GenAI

Work with your AI tool to explore database design approaches